STRIDE Threat Model Analysis Report

AutoDrive Manufacturing - Connected and Autonomous Vehicle Platform

Prepared for: Lab - Threat Modeling

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Executive Summary
AutoDrive Manufacturing operates a safety-critical connected vehicle ecosystem that depends on trusted vehicle identity, secure cloud telemetry, resilient over-the-air firmware delivery, protected messaging, and controlled third-party API access. This assessment applies the STRIDE framework across the full platform and identifies the highest-priority risks as firmware tampering, denial-of-service against cloud authentication and update services, and data exposure through API misuse or misconfiguration. Because compromise in this environment can affect both business operations and vehicle safety, the recommended control strategy emphasizes defense in depth: strong identity and certificate lifecycle management, signed and verified updates, hardened API security, comprehensive logging, and continuous anomaly detection.

1. System Overview and Scope

AutoDrive Manufacturing develops connected and autonomous vehicle technologies that rely on AWS-hosted cloud services to authenticate vehicles, collect telemetry, deliver remote diagnostics, and distribute over-the-air (OTA) firmware updates. The platform also supports secure vehicle-to-cloud messaging and third-party integration through a REST API used by authorized partners and service centers.

Security is mission-critical because system compromise can affect confidentiality of vehicle and customer data, integrity of firmware and operational commands, availability of cloud services, and ultimately vehicle safety. In an automotive context, cybersecurity failures may create operational, legal, regulatory, and safety consequences.

Key components in scope:

- Vehicle Authentication System
- Vehicle Telemetry and Data Management
- OTA Software Update System
- Secure Messaging and Communication System
- Third-Party Integration API

Scope of analysis:

This report analyzes the entire connected vehicle platform at a component level, with deeper attention to authentication, data storage and transport, update distribution, messaging flows, and API exposure because these areas map directly to the recent incidents described by AutoDrive.

2. System Components Breakdown

Component	Description	Security Contribution
Vehicle Authentication System	Authenticates vehicles to cloud services using digital certificates and secure key exchange. Supports MFA for remote diagnostics personnel and password reset for registered fleet users.	Prevents unauthorized platform access and establishes trusted identity for vehicle and user interactions.
Vehicle Telemetry and Data Management	Collects and stores diagnostics, location, and performance data in cloud databases protected by AES-256 at rest and TLS 1.2/1.3 in transit.	Protects confidentiality and integrity of telemetry while enabling authorized monitoring and diagnostics.
OTA Software Update System	Distributes remote firmware updates, validates integrity with cryptographic signatures, and supports rollback if update failure or compromise is detected.	Protects firmware integrity and supports secure patching and recovery.
Secure Messaging and Communication System	Handles encrypted vehicle-to-cloud messages for updates, alerts, and security warnings, with anomaly logging and monitoring.	Maintains trusted command and alert exchange while enabling detection of suspicious behavior.
Third-Party Integration API	Provides OAuth-protected REST endpoints and API key access for diagnostics, log retrieval, and model-specific software update requests, with rate limiting enabled.	Controls partner access, limits abuse, and restricts exposure of high-value operational data and functions.

3. STRIDE Threat Identification

The following matrix identifies realistic threats mapped to the STRIDE model for AutoDrive's in-scope components.

Component	STRIDE Category	Threat Description
Vehicle Authentication System	Spoofing	Attackers use stolen fleet credentials or cloned device certificates to impersonate legitimate users or vehicles.
Vehicle Authentication System	Denial of Service	Credential stuffing and high-volume login attempts overwhelm authentication services and disrupt legitimate vehicle connectivity.
Vehicle Authentication System	Elevation of Privilege	Weak password reset or insufficient role checks allow an attacker to gain engineer or administrator-level access.
Telemetry and Data Management	Information Disclosure	Misconfigured storage, logs, or API responses expose sensitive telemetry such as location, diagnostics, and performance data.
Telemetry and Data Management	Tampering	An attacker modifies stored telemetry or metadata, reducing trust in diagnostics and incident response decisions.
Telemetry and Data Management	Repudiation	Users or insiders deny having accessed or changed telemetry records when audit logs are incomplete or alterable.
OTA Software Update System	Tampering	Malicious actors attempt to alter firmware packages, update manifests, or delivery channels to install unauthorized code.
OTA Software Update System	Spoofing	An attacker impersonates the update server or abuses partner API functions to trigger unauthorized update requests.
OTA Software Update System	Denial of Service	Attackers disrupt update distribution or rollback services, delaying security patches and affecting fleet availability.
Secure Messaging System	Tampering	Messages are modified in transit or replayed to inject false warnings, outdated commands, or unsafe operational signals.
Secure Messaging System	Repudiation	Lack of strong non-repudiation allows parties to deny sending sensitive operational or emergency messages.
Third-Party Integration API	Information Disclosure	API misconfiguration, broken access control, or excessive data exposure leaks logs, software versions, or telemetry.
Third-Party Integration API	Elevation of Privilege	OAuth token abuse, weak authorization checks, or exposed keys allow a partner or attacker to access restricted functions.

4. Risk Assessment and Impact

Risk ratings below reflect practical attack feasibility, current indicators from recent incidents, potential business disruption, and possible downstream vehicle safety impact.

Threat	STRIDE	Likelihood	Impact	Overall Risk
Credential stuffing or certificate spoofing enables unauthorized access	Spoofing	High	High	Critical
API misconfiguration exposes telemetry, logs, or software details	Information Disclosure	Medium	High	High
Firmware tampering introduces malicious or untrusted code	Tampering	Medium	High	Critical
DoS against authentication or update infrastructure disrupts vehicle connectivity	Denial of Service	High	High	Critical
Weak password reset or broken authorization leads to admin/engineer access	Elevation of Privilege	Medium	High	High
Telemetry or logs are altered, reducing integrity of diagnostics and investigations	Tampering / Repudiation	Medium	Medium	Medium
Message replay or manipulation injects false warnings or commands	Tampering	Medium	High	High
Insufficient auditability allows insiders or partners to deny actions	Repudiation	Low	Medium	Medium

5. Mitigation Strategies

Threat	STRIDE	Mitigation Strategy
Credential theft or certificate spoofing	Spoofing	Enforce phishing-resistant MFA for users, mutual TLS for vehicles, certificate revocation, device attestation, and anomaly-based login detection.
API data exposure or abuse	Information Disclosure	Apply least-privilege authorization, secure OAuth scopes, API gateway validation, rate limiting, response minimization, secret rotation, and continuous API security testing.
Firmware tampering	Tampering	Use code signing, secure boot, signed manifests, hardware-backed key protection, isolated build pipelines, update package hashing, and mandatory signature verification before installation.
DoS against auth or OTA services	Denial of Service	Adopt AWS DDoS protections, autoscaling, WAF rules, request throttling, resilient queuing, and fallback update scheduling for critical fleet operations.
Privilege escalation through weak reset or RBAC gaps	Elevation of Privilege	Harden password reset with step-up verification, short-lived reset tokens, strict RBAC/ABAC, privileged access management, and approval workflows for sensitive actions.
Telemetry/log tampering	Tampering / Repudiation	Use immutable logging, write-once storage for critical events, database integrity monitoring, cryptographic checksums, and separation of duties for administrative changes.
Message manipulation or replay	Tampering	Implement message signing, timestamps, nonces, replay protection, certificate-based trust, and strict server-side validation of message origin and freshness.
Action denial due to weak audit trails	Repudiation	Maintain centralized audit logging with synchronized time, signed log streams, retention controls, and monitoring that links user, device, API token, and action context.

6. Summary and Recommendations

Top Three Most Critical Threats

1. **Firmware tampering -** This is a top priority because it directly threatens software integrity in a safety-critical automotive environment. A successful compromise could distribute malicious code to multiple vehicles and create operational or safety consequences.
2. **Denial-of-service against authentication and OTA infrastructure -** This is high priority because AutoDrive has already observed traffic spikes. Service disruption can block legitimate connectivity, delay urgent patches, and affect fleet availability at scale.
3. **API data exposure and abuse -** This is critical because the API exposes high-value operational data and update-related functions. Misconfiguration or weak access control may lead to data leakage, reconnaissance, or unauthorized actions by external parties.

Final Recommendations for Improving Security

- Prioritize identity assurance by strengthening MFA, certificate lifecycle management, and privileged access controls.
- Treat the OTA pipeline as a crown-jewel service and secure it end-to-end with signed builds, secure boot, strict release approvals, and rapid rollback governance.
- Harden API security through least privilege, continuous testing, gateway enforcement, and tighter monitoring for abnormal request patterns.
- Expand centralized monitoring to correlate authentication events, firmware anomalies, messaging irregularities, and cloud infrastructure telemetry in near real time.
- Improve resilience through DDoS protection, autoscaling, rate limiting, and tested incident response playbooks aligned to automotive cyber incident scenarios.
- Preserve forensic integrity with immutable logging, time synchronization, and audit trails that support both operational investigation and regulatory accountability.

Appendix: What the Lab Is Asking You to Do

The lab expects a structured threat model, not a technical exploit exercise. You are expected to explain what the system does, define which components are in scope, identify realistic threats using STRIDE, rate each threat by likelihood and impact, propose mitigations, and then summarize the most serious risks. To score highly, your answers should be specific to AutoDrive's connected vehicle environment and written in a clear, professional report style.